

MATH3026/4022 Lecture 3

Moving Average and Autoregressive Processes

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2nd February 2026

- ▶ Moving average processes
 - ▶ Autocorrelation function (ACF) and other properties
 - ▶ Invertibility
- ▶ The general linear process
- ▶ Causal time series
- ▶ Introduction to autoregressive time series
 - ▶ Autoregressive process of order 1

Introduction

During Lecture 2 we explored important concepts concerning time series, such as stationarity plus mean and autocovariance functions.

We also introduced the **purely random** or **white noise** process and began to consider approaches to modelling time series data.

In this lecture we begin to consider formal modelling for time series.

We begin by introducing a more flexible time series model, known as a **moving average process**.

Moving Average Process

Let $\{Z_t : t \in \mathbb{Z}\}$ be a purely random process (white noise) with variance σ_z^2 .

Then $\{X_t : t \in \mathbb{Z}\}$ is a **moving average process** of order q if

$$X_t = \theta_0 Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}.$$

Here the θ terms are constants and, often, θ_0 is set to be equal to 1.

Usually, we describe the **moving average process** of **order** q as an $MA(q)$ process.

Since $\mathbb{E}(Z_t) = 0$ and $\text{Var}(Z_t) = \sigma_z^2$ for all t , we see that

$$\mathbb{E}(X_t) = 0 \quad \text{and} \quad \text{Var}(X_t) = \sigma_z^2 \sum_{j=0}^q \theta_j^2$$

Moving Average Process

In addition, if $\{X_t\}$ is an $MA(q)$ process, we can show that

$$\text{Cov}(X_t, X_{t+h}) = \gamma(h) = \begin{cases} \sigma_z^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h} & h = 0, 1, \dots, q; \\ 0 & h > q; \\ \gamma(-h) & h < 0. \end{cases}$$

We shall now derive the above result.

Derivation: $\gamma(h)$ for the $MA(q)$ process

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Moving Average Process

We see that the $MA(q)$ process is second-order (weakly) stationary, noting that

- (i) $\mathbb{E}(X_t) = 0$ for all t .
- (ii) $\text{Var}(X_t) = \sigma_z^2 \sum_{i=0}^q \theta_i^2$ for all t .
- (iii) $\gamma(t, t+h)$ does not depend on t and depends only on the lag h .

Moving Average Process

Hence, the autocorrelation function (ACF) for the $MA(q)$ process is

$$\rho(h) = \begin{cases} 1 & h = 0; \\ \frac{\sum_{j=0}^{q-h} \theta_j \theta_{j+h}}{\sum_{j=0}^q \theta_j^2} & h = 0, \dots, q; \\ 0 & h > q; \\ \rho(-h) & h < 0. \end{cases}$$

We see that a feature of an $MA(q)$ process is that the ACF 'cuts off' after lag q .

Invertibility

We note that there are no restrictions on the parameters $\{\theta_i\}$ for the $MA(q)$ process to be stationary.

However, we might like to impose conditions on the parameters $\{\theta_i\}$ to ensure that the process is **invertible**.

For example, consider the following $MA(1)$ process

$$X_t = Z_t + \theta Z_{t-1}$$

Rearranging, we see that

$$\begin{aligned} Z_t &= X_t - \theta Z_{t-1} \\ &= X_t - \theta (X_{t-1} - \theta Z_{t-2}) \quad \text{since } Z_{t-1} = X_{t-1} - \theta Z_{t-2} \\ &= X_t - \theta X_{t-1} + \theta^2 X_{t-2} - \dots \\ &= \sum_{j=0}^{\infty} (-\theta)^j X_{t-j} \end{aligned}$$

We see that the sequence of coefficients $\{(-\theta)^j : j = 0, 1, \dots\}$ (on the previous slide) will converge if $|\theta| < 1$. If this holds, we say that the process is **invertible**.

Definition 3.1: Invertible Process

The process $\{X_t\}$ is **invertible** if the random error at time t , Z_t , may be written in the form

$$Z_t = \sum_{j=0}^{\infty} \alpha_j X_{t-j}$$

where $\{\alpha_j : j = 0, 1, \dots\}$ are coefficients such that $\sum_{j=0}^{\infty} |\alpha_j| < \infty$.

In words, we see that an invertible process is one where the random error at each time point can be written as a sum of previous values of the series.

This type of process is known as an **autoregressive process**. We will cover such processes in more detail during Lecture 4.

If a condition of invertibility is imposed when fitting an MA process then this condition will ensure that there is a unique moving average process for a given autocorrelation function (ACF).

Invertibility and the $MA(q)$ process

An $MA(q)$ process may be written

$$\begin{aligned}X_t &= \sum_{j=0}^q \theta_j Z_{t-j} \\&= (\theta_0 + \theta_1 B + \dots + \theta_q B^q) Z_t \\&= \theta(B) Z_t \text{ where } \theta(B) = \theta_0 + \theta_1 B + \dots + \theta_q B^q.\end{aligned}$$

It can be shown that an $MA(q)$ process is invertible if the roots of the equation

$$\theta(z) = \theta_0 + \theta_1 z + \dots + \theta_q z^q = 0$$

lie outside the unit circle (i.e. each root has an absolute value > 1).

The 'unit circle' is the circle with radius 1.

$MA(1)$ process

The $MA(1)$ process (with $\theta_0 = 1$) is the simplest example of a moving average process, where

$$X_t = Z_t + \theta_1 Z_{t-1}$$

and the ACF is

$$\rho(h) = \begin{cases} 1 & h = 0; \\ \frac{\theta_1}{1 + \theta_1^2} & h \in \{-1, +1\}; \\ 0 & \text{otherwise.} \end{cases}$$

Example 3.1

For what values of β_1 will the following $MA(1)$ process be invertible?

$$X_t = Z_t + \beta_1 Z_{t-1}.$$

The General Linear Process

We define an important stochastic process, known as a **general linear process**, that is used in the analysis of many common time series models.

Definition 3.2: General Linear Process

Suppose that $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with variance σ_z^2 . Then the **general linear process** $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = \sum_{j=0}^{\infty} \alpha_j Z_{t-j}$$

where $\{\alpha_j : j = 0, 1, \dots\}$ are appropriate coefficients.

The General Linear Process

We see that the general linear process is a **moving average process**, possibly of infinite order.

Often, we might refer to such a process as an $MA(\infty)$ process (or describe a process as written in $MA(\infty)$ form).

We shall consider properties of the general linear process that would ensure that the process is 'well behaved'.

It is easy to see that

$$\mathbb{E}(X_t) = \sum_{j=0}^{\infty} \alpha_j \mathbb{E}(Z_{t-j}) = 0$$

since $\{Z_t\}$ is a white noise process.

So the process $\{X_t\}$ will have mean zero regardless of the values of the α_j s.

General Linear Process

The variance of X_t is

$$\begin{aligned}\text{Var}(X_t) &= \text{Var}\left(\sum_{j=0}^{\infty} \alpha_j Z_{t-j}\right) \\ &= \sum_{j=0}^{\infty} \alpha_j^2 \text{Var}(Z_{t-j}) \quad (\text{independence of } Z_t\text{'s}) \\ &= \sigma_z^2 \sum_{j=0}^{\infty} \alpha_j^2\end{aligned}$$

Hence, we see that we must have $\sum_{j=0}^{\infty} \alpha_j^2 < \infty$ for the variance of the process to be finite.

General Linear Process

The covariance of X_t and X_{t+h} is

$$\begin{aligned}\text{Cov}(X_t, X_{t+h}) &= \mathbb{E}(X_t X_{t+h}) - \mathbb{E}(X_t)\mathbb{E}(X_{t+h}) \\ &= \mathbb{E}(X_t X_{t+h}) \quad \text{since } \mathbb{E}(X_t) = 0 \text{ for all } t. \\ &= \mathbb{E} \left[\left(\sum_{j=0}^{\infty} \alpha_j Z_{t-j} \sum_{k=0}^{\infty} \alpha_k Z_{t+h-k} \right) \right] \\ &= \sum_j \sum_k \alpha_j \alpha_k \mathbb{E}(Z_{t-j} Z_{t+h-k}).\end{aligned}$$

We note that

$$\mathbb{E}(Z_{t-j} Z_{t+h-k}) = \begin{cases} \mathbb{E}(Z_{t-j}^2) = \sigma_z^2 & j = -h + k; \\ 0 & \text{otherwise.} \end{cases}$$

General Linear Process

Hence, the sum $\sum_j \sum_k \alpha_j \alpha_k \mathbb{E}(Z_{t-j} Z_{t+h-k})$ is non-zero only where

$$j = -h + k \implies k = j + h.$$

Hence

$$\text{Cov}(X_t, X_{t+h}) = \gamma(h) = \sigma_z^2 \sum_{j=0}^{\infty} \alpha_j \alpha_{j+h}$$

We note that $\gamma(h)$ does not depend on t and so the general linear process is (weakly) stationary (subject to the constraint $\sum_{j=0}^{\infty} \alpha_j^2 < \infty$).

If $h > 0$ the ACF is given by

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \frac{\sum_{j=0}^{\infty} \alpha_j \alpha_{j+h}}{\sum_{j=0}^{\infty} \alpha_j^2}.$$

Another method to assess whether or not the general linear process is 'well-behaved' is to see if the process **converges in mean square** to a random variable

A definition of convergence in mean square is provided below.

Definition 3.3: Convergence in Mean Square

Suppose that $\{Y_n\}_{n=1}^{\infty}$ are a sequence of random variables. We say that Y_n converges in mean square to the random variable Y if

$$\mathbb{E} \left[(Y_n - Y)^2 \right] \rightarrow 0 \text{ as } n \rightarrow \infty.$$

General Linear Process

We define

$$X_{tn} = \sum_{j=0}^n \alpha_j Z_{t-j}$$

and we shall explore whether or not X_{tn} converges in mean square to X_t . In other words, we ask the question:

$$\text{Is } \lim_{n \rightarrow \infty} \mathbb{E} [(X_{tn} - X_t)^2] = 0?$$

We will show that this is true if

$$\sum_{j=0}^{\infty} \alpha_j^2 < \infty.$$

This is desirable because this is also a condition for (weak) stationarity.

General Linear Process - Convergence in Mean Square

General Linear Process - Convergence in Mean Square

Further Assumptions/Conditions

A further assumption we shall make is that past values of the general linear process do not influence the current value of the white noise.

That is, that X_s and Z_t are uncorrelated where $s < t$. We write this as

$$\mathbb{E}(X_s Z_t) = 0 \quad \text{where } s < t.$$

Another condition that we would like to impose is

$$\sum_{j=0}^{\infty} |\alpha_j| < \infty.$$

This condition ensures (weak) stationarity but also yields a property known as **causality**.

We say that X_t is a **causal function** of **current** and **past** white noise and note that X_t does not depend on future white noise.

Autoregressive Processes

At this point, we have defined a general linear process and considered various properties of this process.

Notably, we have seen that the general linear process is a moving average process of (possibly) infinite order.

Now we turn our attention to another common type of time series, known as an **autoregressive process**.

Autoregressive Processes

Let $\{Z_t : t \in \mathbb{Z}\}$ be a purely random process (white noise) with variance σ_z^2 .

Then $\{X_t : t \in \mathbb{Z}\}$ is an **autoregressive process** of order p if

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + Z_t.$$

Here, the ϕ terms are constant and, usually, we describe the autoregressive process of order p as an $AR(p)$ process.

In words, the current value of the process is a function of the past p values of the process, plus a random error term.

AR(1) Process and the General Linear Process

Consider the following $AR(1)$ process

$$X_t = \phi X_{t-1} + Z_t$$

Clearly, we see that

$$\begin{aligned} X_t &= Z_t + \phi(Z_{t-1} + \phi X_{t-2}) \quad (\text{as } X_{t-1} = \phi X_{t-2} + Z_{t-1}) \\ &= Z_t + \phi Z_{t-1} + \phi^2(Z_{t-2} + \phi X_{t-3}) \\ &= \dots \\ &= \sum_{j=0}^{\infty} \phi^j Z_{t-j}. \end{aligned}$$

The $AR(1)$ process can be re-written as a general linear process (an $MA(\infty)$ process).

We note that when $|\phi| < 1$ we see that $\sum_{j=0}^{\infty} \phi^j$ will converge.

AR(1) Process

Before exploring the general $AR(p)$ process in more detail (in Lecture 4), we'll explore properties of the simpler $AR(1)$ process as this may provide some useful insight into the behaviour of more general $AR(p)$ processes.

The $AR(1)$ process is written

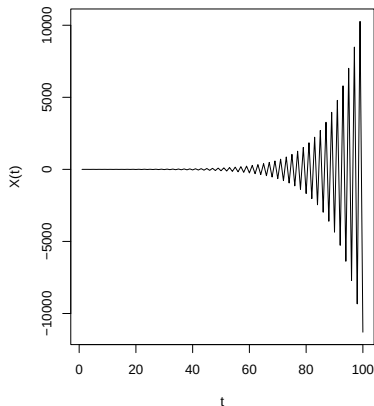
$$X_t = \phi X_{t-1} + Z_t.$$

Let us consider values of ϕ for which the $AR(1)$ process may be stationary.

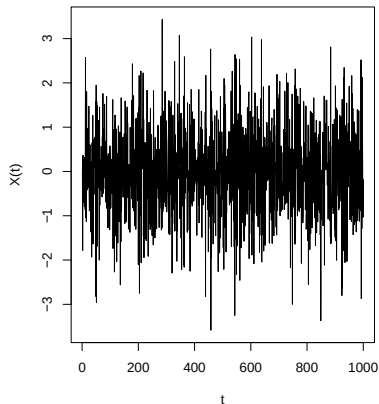
The next two slides show plots of simulated $AR(1)$ processes for different values of ϕ .

AR(1) Models

AR(1) process, $\phi = -1.1$

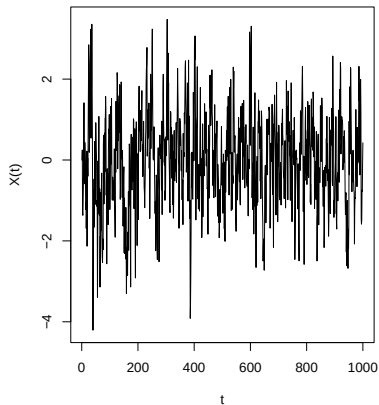


AR(1) process, $\phi = -0.5$

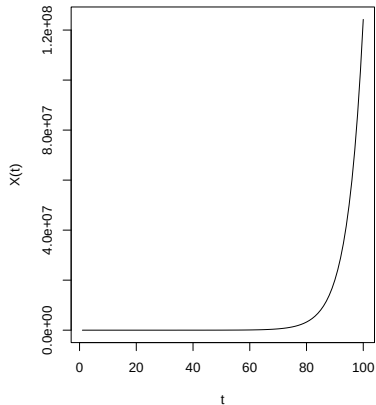


AR(1) Models

AR(1) process, $\phi = 0.6$



AR(1) process, $\phi = 1.2$



AR(1) Models

We see that, where $\phi = -1.1$, as time increases the process increases in magnitude, fluctuating between large positive and large negative values.

Where $\phi = 1.2$, the process increases greatly with time (asymptotically to infinity).

In contrast, where $\phi = -0.5$ and where $\phi = 0.6$ the process appears stable and, perhaps, stationary.

Perhaps something interesting occurs at $\phi = \pm 1$?

We shall investigate this further, looking at the mean, variance and autocovariance for $|\phi| < 1$, $|\phi| > 1$ and $|\phi| = 1$.

AR(1) Model: Stationarity

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AR(1) Model: Stationarity

AR(1) Process - Summary

To summarise, for the $AR(1)$ process

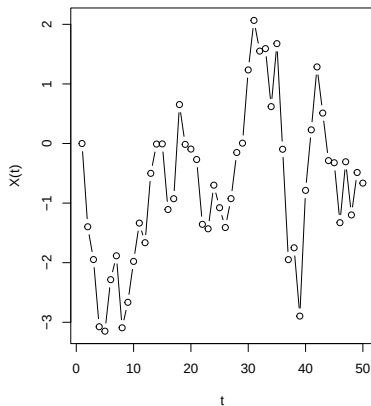
$$X_t = \phi X_{t-1} + Z_t.$$

The process is stationary if $|\phi| < 1$ and, in this case, the autocorrelation function is

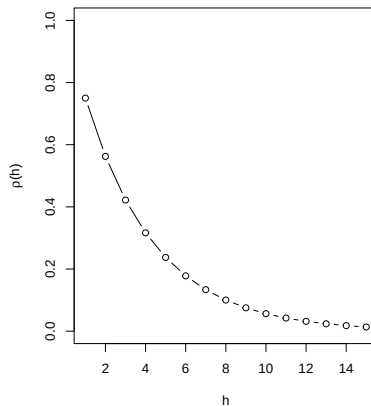
$$\begin{aligned}\rho(h) &= \frac{\gamma(h)}{\gamma(0)} \\ &= \frac{\gamma(h)}{\text{Var}(X_t)} \\ &= \frac{\sigma_z^2 \phi^{|h|} / (1 - \phi^2)}{\sigma_z^2 / (1 - \phi^2)} \\ &= \phi^{|h|}.\end{aligned}$$

Example $AR(1)$ process and ACF where $\phi = 0.75$

AR(1) process, $\phi = 0.75$

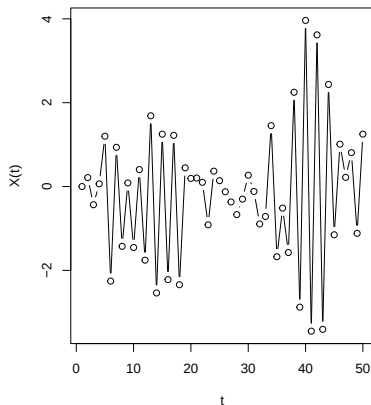


ACF: AR(1) process, $\phi = 0.75$

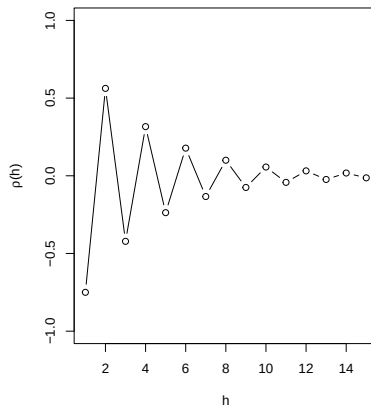


Example $AR(1)$ process and ACF where $\phi = -0.75$

AR(1) process, $\phi = -0.75$



ACF: AR(1) process, $\phi = -0.75$



Yule-Walker Equations for an $AR(1)$ process

If we assume, a priori, that an $AR(1)$ process is stationary (whence $\mathbb{E}(X_t) = \mu = 0$) then we can take the equation

$$X_t = \phi X_{t-1} + Z_t$$

and, multiplying through by X_{t-h} and taking expectations, we see

$$\begin{aligned}\mathbb{E}(X_t X_{t-h}) &= \phi \mathbb{E}(X_{t-1} X_{t-h}) + \mathbb{E}(Z_t X_{t-h}) \\ \implies \gamma(-h) &= \phi \gamma(-h+1) \quad (\text{since } \mathbb{E}(Z_t X_{t-h}) = 0)\end{aligned}$$

Since $\gamma(h) = \gamma(-h)$ (and dividing through by $\gamma(0)$) we obtain the set of difference equations

$$\rho(h) - \phi \rho(h-1) = 0.$$

These are known as **Yule-Walker equations** and have solution

$$\rho(h) = \phi^{|h|}.$$

Autoregressive Process

In general, an autoregressive process can be written as a moving average process.

Compare this with the result seen in Lecture 2 where we saw that a moving average process can be written as the infinite sum of past values of the process (i.e. an autoregressive process).

This is a useful property of both MA and AR processes.

Lecture 3: Learning Outcomes

- ▶ Know what is meant by a **moving average process** and be able to derive and use the autocorrelation/autocovariance function of the $MA(q)$ process.
- ▶ Understand what is meant by an **invertible** process and be able to assess and derive conditions for invertibility for moving average processes.
- ▶ Understand the definition of a **general linear process** and be able to derive and use the autocorrelation/autocovariance function of the $MA(q)$ process.
- ▶ Know what is meant by an **autoregressive process** and be able to explore various properties of the $AR(1)$ process.